Actual first-collision obstruction for fixed supports

For every deterministic labeling h:Z/p→Fin r, let f(x)=x+h(x)+1. Lean defines T as the least index at which the actual iteration starting at0 repeats an earlier state. It proves p≤rT and T≤p. Under the actual independent uniform node-label product PMF, p≤rE[T] and E[T]≤p. [pcc:unverified]

With r=20 and prime p>20, all increments are distinct and nonzero and actual indegree has mean1, variance19/20. Yet no real C bounds $E[T]^2$ by Cp uniformly over these primes. [pcc:unverified]

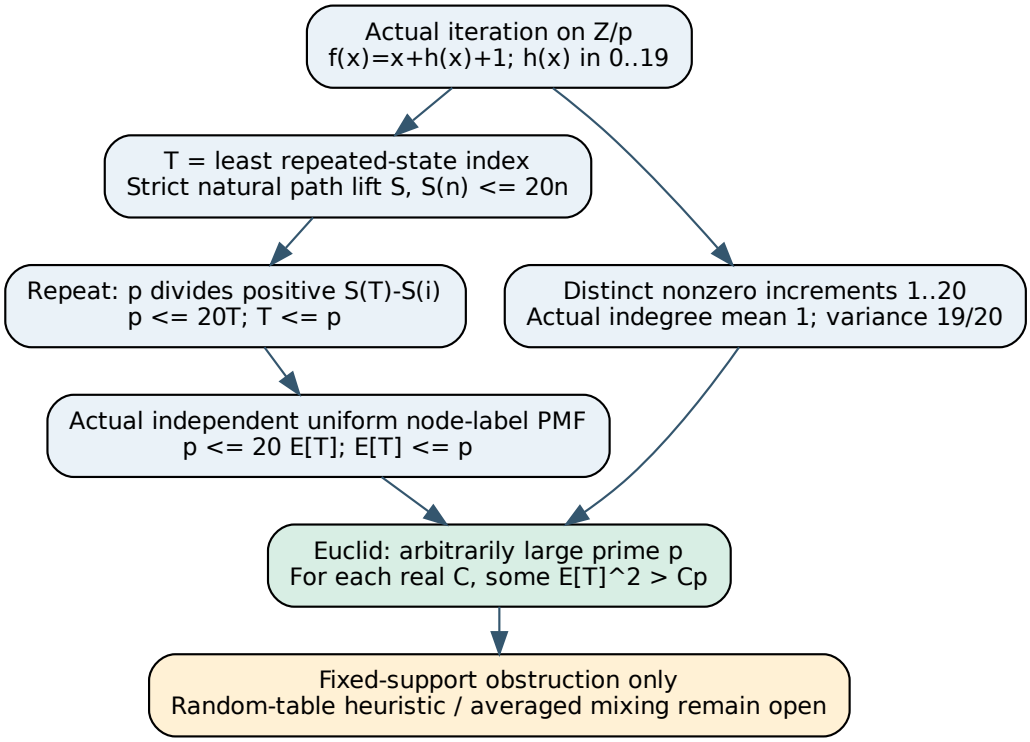

Figure 1: Actual iteration, normalized probability and prime-infinitude proof. [pcc:unverified]

Proof obligations, source qualifications and assurance

The proof lifts each actual modular path to a strictly increasing natural path S. Its increments lie in1..r, so S(n)≤rn. Repeated residues at i<T imply p divides S(T)−S(i)>0, giving p≤rT. A pigeonhole repeat bound gives T≤p. No collision-time model is substituted for the actual iteration. [pcc:unverified]

Monotonicity of the finite expectation transfers these pointwise bounds. The weights are proved nonnegative and normalized, and the expectation is proved equal to the genuine product PMF sum. The frozen parent proves the actual edge-count variance for every distinct increment table. At r20 this is19/20. Mathlib's proved Euclid theorem supplies prime p>400C, contradicting $E[T]^2 \le Cp$ and p≤20E[T]. An Archimedean bound extends the disproof from natural to real C. [pcc:unverified]

Declaration	Obligation
firstCollision_spec / firstCollision_le	Actual least repeated state and upper bound
repeat_lower_bound	Strict natural lift, positive displacement divisible by p
expectation_eq_pmf	Real expectation equals actual PMF sum
meanCollision_lower / upper	Linear lower bound and finite upper bound
twenty_partition_certificate	Actual support, iteration, PMF and moments
arbitrarily_large_collision_ratio	Prime-modulus witnesses for every natural coefficient
universal_variance_only_collision_bound_false	Direct negation for every real coefficient

Galbraith's14.2.1 definition samples the increment table randomly. This study fixes the special support1..20. Hildebrand1994 Theorem1 averages over random distinct supports; its fixed-support distinction is explicit. These source assumptions prevent promoting this counterfamily to a refutation of their qualified random-support statements. Relevant primary sections and exact hashes are in evidence/source-review.json. This is formalization of a known fixed-support obstruction; no novelty claim. [pcc:unverified]

Lean4.35.0 / Mathlib56a00585 are pinned. Only propext, Classical.choice and Quot.sound; no holes, custom axioms, native_decide, waived linters or increased elaboration settings. Issued parent proofs are unchanged. Full gate, source pins, failed attempts and actual PCC replay/standing are recorded separately. Signed standing remains unverified. [pcc:unverified]

No native experiment, sampling, timing or paid compute. KB MCP absent, wrapper127; bounded primary/repository/Mathlib fallback. Conductor and lean4checker unavailable. Proof map: README.md. Exact dependency pins: sources.json. Actual local verdict and signed standing: verification.json. [pcc:unverified]